

Pet Adoption Center

Start with a list of hard code of animals

4 classes

- **Animal Class - all classes is-A animal (SuperClass)**
- (static variables for age/weight etc)
 - Bird object
 - ID(int), weight, species(Str) (use If statements for age/weight ex 1-3 = young age, 4-7 = middle age between this many lbs = light, average and heavy.
 - Reptile Object
 - ID(int), weight species (Str)
 - Mammal Object
 - PurchaseAnimalmethod()
 - Prompt (Bird, Reptile, Mammal)
 - Prompt (old, young, middle aged)
 - Prompt (heavy, light, average)
 - Return list of animals
- Override method super class print statement given different parameters.
- animalID = animalCount++ (int)
- Stores name(str), age(int), type of animal(str), weight (double), Make these variables static
- Use **hashmap** to link name with ID
- **Arrays** for looping purposes
- AddAnimal method overload
 - addAnimal(String AnimalType)
 - addAnimal(String AnimalType, Int Age)
 - addAnimal(String Animal Type, Int Age, Double Weight)
- DeterminePriceMethod(age,weight)
- For add Animal, do either exchangeforAnimal or exchangeforPrice
- Prompt user for animal info and then displays either price or

Buy option method

- four options Reptile, mammal, bird, see all options
 - Then selected one (reptile)
 - Shows all options on animals etc
 - Prompt selection of animal and return price
 - Confirm sale

Sell option method

- Enter what type of animal you are selling
- Input price calculator bird info(age and weight)
- Then the system would give you a price range within this (you can deny or accept)
- If accept, prompt for additional information (class specific)

Exchange option method

- enter what type of animal you are exchanging and what do you want

- displays animal within the same price range of the animal being exchange (price match)
- make new animal object out of inputted data
- Bird
 - Is a bird
 - Store birds as `arrayList` of strings (loops plus ++)
- Reptile
 - Lizard is a reptile
 - `HashSet <String>`
- Mammal
 - Dog is a mammal
 - `HashMap <String,Integer> <mammalType,Age>`
- Use a `Hashset<String>` to add any new animals that are brought in
- Hashmap to add or call animal ID and Name
- Mammal, reptile, and bird all has-A ID
- arrayList to store mammals, reptiles, and birds by weight
 - For loop / if else to store animals objects by age
- Maybe within mammal and reptile class, arrayList / array of the different types ex
 - Dog arrayList / array
 - Cat Arraylist / array
 - Lizard arraylist / array
 - Snake arraylist / array etc

FINALCODE

New project FinalProject
Package petAdoption
Class is Animal

```
package petAdoption;
import java.util.Scanner;
public class Animal {
    Scanner scnr = new Scanner(System.in);
    private static int numAnimals;
    protected int id;
    protected int age;
    protected double weight;
    protected String species;
    protected int price;
    public Animal() {
        id = numAnimals;
        age = 0;
        weight = 0.0;
        species = "TBD";
        numAnimals++;
    }
    public int getPrice() {
        return price;
    }
    public void setPrice(int price) {
        this.price = price;
    }
    public int getId() {
        return id;
    }
    public void setId(int id) {
        this.id = numAnimals;
    }
    public static int getNumAnimals() {
        return numAnimals;
    }
    public static void setNumAnimals(int numAnimals) {
        Animal.numAnimals = numAnimals;
    }
}
```

```

    }
    public int getAge() {
        return age;
    }
    public void setAge(int age) {
        this.age = age;
    }
    public double getWeight() {
        return weight;
    }
    public void setWeight(double weight) {
        this.weight = weight;
    }
    public String getSpecies() {
        return species;
    }
    public void setSpecies(String species) {
        this.species = species;
    }
    public Animal(String s, int a, double w) {
        id = numAnimals;
        species = s;
        age = a;
        weight = w;
        numAnimals++;
    }
    public void printInfo(Object Animal) {
        System.out.println("ID: " + this.getId());
        System.out.println("Species: " + this.getSpecies());
        System.out.println("Age: " + this.getAge());
        System.out.println("Weight: " + this.getWeight() + " lbs");
        System.out.println("Price: " + this.getPrice());
    }
}

```

BIRD CLASS

```
package petAdoption;
```

```
import java.util.ArrayList;
public class Bird extends Animal {
    private String birdType;
    private String feathColor;
    public Bird() {
        super();
        species = "Bird";
        birdType = "TBD";
        feathColor = "blank";
        price = 25;
    }
    @Override
    public int getPrice() {
        return price;
    }
    @Override
    public void setPrice(int price) {
        this.price = price;
    }
    public String getBirdType() {
        return birdType;
    }
    public void setBirdType(String birdType) {
        this.birdType = birdType;
    }
    public String getFeathColor() {
        return feathColor;
    }
    public void setFeathColor(String feathColor) {
        this.feathColor = feathColor;
    }
    public Bird(int a, double w) {
        age = a;
        weight = w;
    }
    public Bird(String bt, int a, double w, String fc) {
        birdType = bt;
        age = a;
        weight = w;
        feathColor = fc;
    }
}
```

```

        species = "Bird";
        price = 25;
    }

    public static void birdBuy(ArrayList<Bird> birdStock) {
        System.out.println("Here is our list of Birds:");
        for (Bird b : birdStock) {
            b.printInfo(b);
        }
        System.out.println("What Bird type would you like to purchase?");
    }

    public static void birdPrint(ArrayList<Bird> birdStock) {
        System.out.println("Here is our list of Birds:");
        for (Bird b : birdStock) {
            b.printInfo(b);
        }
    }
}

@Override
public void printInfo(Object Bird) {
    System.out.println("Type: " + this.getBirdType());
    super.printInfo(Bird);
    System.out.println("Feather color: " + this.getFeathColor());
}
}

```

MAMMAL

```

package petAdoption;
import java.util.HashSet;
public class Mammal extends Animal {
    private String mammalType;
    private String furColor;
    public Mammal() {
        super();
        species = "Mammal";
        mammalType = "TBD";
        furColor = "blank";
        price = 100;
    }
    @Override
    public int getPrice() {

```

```

        return price;
    }
    @Override
    public void setPrice(int price) {
        this.price = price;
    }
    public String getMammalType() {
        return mammalType;
    }
    public void setMammalType(String mammalType) {
        this.mammalType = mammalType;
    }
    public String getFurColor() {
        return furColor;
    }
    public void setFurColor(String furColor) {
        this.furColor = furColor;
    }
    public Mammal(String mt, int a, double w, String fc) {
        mammalType = mt;
        age = a;
        weight = w;
        furColor = fc;
        species = "Mammal";
        price = 100;
    }
    public static void mammalBuy(HashSet<Mammal> mammalStock) {
        System.out.println("Here is our list of Mammals:");
        for (Mammal m : mammalStock) {
            m.printInfo(m);
        }
        System.out.println("What animal type would you like to purchase?");
    }
    public static void mammalPrint(HashSet<Mammal> mammalStock) {
        System.out.println("Here is our list of Mammals:");
        for (Mammal m : mammalStock) {
            m.printInfo(m);
        }
    }
    @Override

```

```

        public void printInfo(Object Mammal) {
            System.out.println("Type: " + this.getMammalType());
            super.printInfo(Mammal);
            System.out.println("Fur: " + this.getFurColor());
        }
    }
}

```

REPTILE

```

package petAdoption;
import java.util.ArrayList;
public class Reptile extends Animal {
    private String reptileType;
    private String scalesColor;
    public Reptile() {
        super();
        species = "Reptile";
        reptileType = "TBD";
        scalesColor = "blank";
        price = 50;
    }
    @Override
    public int getPrice() {
        return price;
    }
    @Override
    public void setPrice(int price) {
        this.price = price;
    }
    public String getReptileType() {
        return reptileType;
    }
    public void setReptileType(String reptileType) {
        this.reptileType = reptileType;
    }
    public String getScalesColor() {
        return scalesColor;
    }
}

```



```

    }
    public void setScalesColor(String color) {
        this.scalesColor = color;
    }
    public Reptile(String rt, int a, double w, String sc) {
        reptileType = rt;
        age = a;
        weight = w;
        scalesColor = sc;
        species = "Reptile";
        price = 50;
    }
    public static void reptileBuy(ArrayList<Reptile> reptileStock) {
        System.out.println("Here is our list of Reptiles:");
        for (Reptile r : reptileStock) {
            r.printInfo(r);
        }
        System.out.println("What Reptile type would you like to purchase?");
    }
    public static void reptilePrint(ArrayList<Reptile> reptileStock) {
        System.out.println("Here is our list of Reptiles:");
        for (Reptile r : reptileStock) {
            r.printInfo(r);
        }
    }
}
@Override
public void printInfo(Object Reptile) {
    System.out.println("Type: " + this.getReptileType());
    super.printInfo(Reptile);
    System.out.println("Scales: " + this.getScalesColor());
}
}

```

Shop

```
package petAdoption;
```

```
import java.util.ArrayList;
import java.util.HashMap;
import java.util.HashSet;
import java.util.Iterator;
import java.util.Scanner;

public class Shop {
    Scanner scnr = new Scanner(System.in);
    ArrayList<Bird> birdStock = new ArrayList<>();
    HashSet<Mammal> mammalStock = new HashSet<>();
    ArrayList<Reptile> reptileStock = new ArrayList<>();
    HashMap<Integer, Animal> animalCount = new HashMap<>();
    public int credit = 0;

    public int getCredit() {
        return credit;
    }

    public void setCredit(int credit) {
        this.credit = credit;
    }

    public void populateBird() {
        Bird bird1 = new Bird("Parrot", 4, .88, "orange");
        Bird bird2 = new Bird("Parakeet", 2, 1.81, "yellow");
        Bird bird3 = new Bird("Bluejay", 1, 2.31, "blue");
        Bird bird4 = new Bird("Cardinal", 3, .66, "red");
        birdStock.add(bird1);
        birdStock.add(bird2);
        birdStock.add(bird3);
        birdStock.add(bird4);
        animalCount.put(bird1.getId(), bird1);
        animalCount.put(bird2.getId(), bird2);
        animalCount.put(bird3.getId(), bird3);
        animalCount.put(bird4.getId(), bird4);
    }

    public void populateMammal() {
        Mammal mammal1 = new Mammal("Dog", 4, 23, "White");
        Mammal mammal2 = new Mammal("Cat", 3, 11.3, "Orange");
    }
}
```

```

Mammal mammal3 = new Mammal("Gerbil", 1, 6.8, "Black");
Mammal mammal4 = new Mammal("Hamster", 4, 9.4, "Brown");
mammalStock.add(mammal1);
mammalStock.add(mammal2);
mammalStock.add(mammal3);
mammalStock.add(mammal4);
animalCount.put(mammal1.getId(), mammal1);
animalCount.put(mammal2.getId(), mammal2);
animalCount.put(mammal3.getId(), mammal3);
animalCount.put(mammal4.getId(), mammal4);
}

public void populateReptile() {
    Reptile reptile1 = new Reptile("Gecko", 1, .45, "tan");
    Reptile reptile2 = new Reptile("Komodo", 6, 183, "dark green");
    Reptile reptile3 = new Reptile("Lizard", 2, .45, "blue");
    Reptile reptile4 = new Reptile("Snake", 3, 15.3, "tan");
    reptileStock.add(reptile1);
    reptileStock.add(reptile2);
    reptileStock.add(reptile3);
    reptileStock.add(reptile4);
    animalCount.put(reptile1.getId(), reptile1);
    animalCount.put(reptile2.getId(), reptile2);
    animalCount.put(reptile3.getId(), reptile3);
    animalCount.put(reptile4.getId(), reptile4);
}

public void initiateAdopt(Scanner scnr) {
    System.out.println("Choose a pet type:");
    System.out.println("1. Reptile");
    System.out.println("2. Bird");
    System.out.println("3. Mammal");
    int choice = scnr.nextInt();
    if (choice == 3) {
        Mammal.mammalBuy(mammalStock);
        String chosenMammal = scnr.next().trim();
        Iterator<Mammal> itr = mammalStock.iterator();
        while (itr.hasNext()) {
            Mammal tempMam = itr.next();

```

```

        if
(tempMam.getMammalType().equalsIgnoreCase(chosenMammal)) {
            System.out.println("The price of this mammal is $" +
tempMam.getPrice());

            System.out.println("Do you wish to purchase?");
            String response = scnr.next();
            if (!response.equalsIgnoreCase("yes")) {
                System.out.println("Sorry, please come again");
                break;
            } else {
                itr.remove();
                animalCount.remove(tempMam.getId());

Animal.setNumAnimals(Animal.getNumAnimals() - 1);
                System.out.println("Thank you for your
business!");
            }
        }

    }

} else if (choice == 2) {
    Bird.birdBuy(birdStock);
    String chosenBird = scnr.next().trim();
    Iterator<Bird> itr = birdStock.iterator();
    while (itr.hasNext()) {
        Bird tempBird = itr.next();
        if (tempBird.getBirdType().equalsIgnoreCase(chosenBird)) {
            System.out.println("The price of this bird is $" +
tempBird.getPrice());

            System.out.println("Do you wish to purchase?");
            String response = scnr.next();
            if (!response.equalsIgnoreCase("yes")) {
                System.out.println("Sorry, please come again");
                break;
            } else {
                itr.remove();
                animalCount.remove(tempBird.getId());

Animal.setNumAnimals(Animal.getNumAnimals() - 1);

```

```

        System.out.println("Thank you for your
business!");
    }
}

    }
} else if (choice == 1) {
    Reptile.reptileBuy(reptileStock);
    String chosenReptile = scnr.next().trim();
    Iterator<Reptile> itr = reptileStock.iterator();
    while (itr.hasNext()) {
        Reptile tempReptile = itr.next();
        if
(tempReptile.getReptileType().equalsIgnoreCase(chosenReptile)) {
            System.out.println("The price of this reptile is $" +
tempReptile.getPrice());

            System.out.println("Do you wish to purchase?");
            String response = scnr.next();
            if (!response.equalsIgnoreCase("yes")) {
                System.out.println("Sorry, please come again");
                break;
            } else {
                itr.remove();
                animalCount.remove(tempReptile.getId());

Animal.setNumAnimals(Animal.getNumAnimals() - 1);
                System.out.println("Thank you for your
business!");
            }
        }
    }
}

    }
}

    }

}

    }

    public void initiateSell(Scanner scnr) {
        System.out.println("What animal would you like to sell?");
        System.out.println("1. Reptile");
        System.out.println("2. Bird");
    }
}

```

```

System.out.println("3. Mammal");
int choice = scnr.nextInt();
if (choice == 3) {
    System.out.println("What type of mammal are you bringing in?");
    String mamType = scnr.next();
    System.out.println("We can offer you the price of $100");
    System.out.println("Accept or Deny:");
    String response = scnr.next();
    if (!response.equalsIgnoreCase("accept")) {
        System.out.println("Sorry, please come again");
    } else {
        Mammal mammal5 = new Mammal();
        System.out.println("What is the age, weight, and color of
your mammal?");

        int mAge = scnr.nextInt();
        int mWeight = scnr.nextInt();
        String mColor = scnr.next();
        mammal5.setMammalType(mamType);
        mammal5.setAge(mAge);
        mammal5.setWeight(mWeight);
        mammal5.setFurColor(mColor);
        mammal5.setPrice(100);
        mammal5.setId(Animal.getNumAnimals());
        mammalStock.add(mammal5);
        animalCount.put(mammal5.getId(), mammal5);
        System.out.println("Thank you for your business!");
    } // else
} else if (choice == 2) {
    System.out.println("What type of bird are you bringing in?");
    String bType = scnr.next();
    System.out.println("We can offer you the price $25");
    System.out.println("Accept or Deny:");
    String response = scnr.next();
    if (!response.equalsIgnoreCase("accept")) {
        System.out.println("Sorry, please come again");
    } else {
        Bird bird5 = new Bird();
        System.out.println("What is the age, weight, and feather
color of your bird?");

        int bAge = scnr.nextInt();

```

```

        int bWeight = scnr.nextInt();
        String bColor = scnr.next();
        bird5.setBirdType(bType);
        bird5.setAge(bAge);
        bird5.setWeight(bWeight);
        bird5.setFeathColor(bColor);
        bird5.setPrice(25);
        bird5.setId(Animal.getNumAnimals());
        birdStock.add(bird5);
        animalCount.put(bird5.getId(), bird5);
        System.out.println("Thank you for your business!");
    }
} else if (choice == 1) {
    System.out.println("What type of reptile are you bringing in?");
    String repType = scnr.next();
    System.out.println("We can offer you the price $50");
    System.out.println("Accept or Deny:");
    String response = scnr.next();
    if (!response.equalsIgnoreCase("accept")) {
        System.out.println("Sorry, please come again");
    } else {
        Reptile reptile5 = new Reptile();
        System.out.println("What is the age, weight, and scale color
of your reptile?");

        int rAge = scnr.nextInt();
        int rWeight = scnr.nextInt();
        String rColor = scnr.next();
        reptile5.setReptileType(repType);
        reptile5.setAge(rAge);
        reptile5.setWeight(rWeight);
        reptile5.setScalesColor(rColor);
        reptile5.setPrice(50);
        reptile5.setId(Animal.getNumAnimals());
        reptileStock.add(reptile5);
        animalCount.put(reptile5.getId(), reptile5);
        System.out.println("Thank you for your business!");
    }
}
}
} // method

```

```

public void initiateExchange(Scanner scnr) {
    System.out.println("What type of animal are you bringing in?");
    String animalTypes[] = { "Reptile", "Bird", "Mammal" };
    for (String animal : animalTypes) {
        System.out.println(animal);
    }
    String response = scnr.next();
    if (response.equalsIgnoreCase("Mammal")) {
        System.out.println("What type of mammal are you bringing in?");
        String mamType = scnr.next();
        Mammal mammal5 = new Mammal();
        System.out.println("What is the age, weight, and color of your
mammal?");
        int mAge = scnr.nextInt();
        double mWeight = scnr.nextDouble();
        String mColor = scnr.next();
        mammal5.setMammalType(mamType);
        mammal5.setAge(mAge);
        mammal5.setWeight(mWeight);
        mammal5.setFurColor(mColor);
        mammal5.setPrice(100);
        mammal5.setId(Animal.getNumAnimals());
        mammalStock.add(mammal5);
        animalCount.put(mammal5.getId(), mammal5);
        System.out.println("Mammals are worth $100");
        System.out.println("Woud you like to exchange for:");
        System.out.println("1. Mammal and no credit");
        System.out.println("2. Reptile and $50 credit");
        System.out.println("3. Bird and $75 credit");
        int response2 = scnr.nextInt();
        if (response2 == 1) {
            Mammal.mammalBuy(mammalStock);
            String chosenMammal = scnr.next().trim();
            Iterator<Mammal> itr = mammalStock.iterator();
            while (itr.hasNext()) {
                Mammal tempMam = itr.next();
                if
(tempMam.getMammalType().equalsIgnoreCase(chosenMammal)) {
                    itr.remove();
                    animalCount.remove(tempMam.getId());
                }
            }
        }
    }
}

```



```

shopCred.setCredit(50);
System.out.println("You now have a $" +
shopCred.getCredit() + " credit");
System.out.println("Thank you for your
business!");
    }
    }
} else if (response.equalsIgnoreCase("Bird")) {
    System.out.println("What type of bird are you bringing in?");
    String birdType = scnr.next();
    Bird bird5 = new Bird();
    System.out.println("What is the age, weight, and color of your
bird?");

    int bAge = scnr.nextInt();
    double bWeight = scnr.nextDouble();
    String bColor = scnr.next();
    bird5.setBirdType(birdType);
    bird5.setAge(bAge);
    bird5.setWeight(bWeight);
    bird5.setFeathColor(bColor);
    bird5.setPrice(100);
    bird5.setId(Animal.getNumAnimals());
    birdStock.add(bird5);
    animalCount.put(bird5.getId(), bird5);
    System.out.println("Birds are worth $25");
    System.out.println("Woud you like to exchange for:");
    System.out.println("1. Mammal and pay $75");
    System.out.println("2. Reptile and pay $25");
    System.out.println("3. Bird and pay/receive no credit");
    int response2 = scnr.nextInt();
    if (response2 == 1) {
        Mammal.mammalBuy(mammalStock);
        String chosenMammal = scnr.next().trim();
        Iterator<Mammal> itr = mammalStock.iterator();
        while (itr.hasNext()) {
            Mammal tempMam = itr.next();
            if
(tempMam.getMammalType().equalsIgnoreCase(chosenMammal)) {
                itr.remove();

```

```

        animalCount.remove(tempMam.getId());

Animal.setNumAnimals(Animal.getNumAnimals() - 1);
        Shop shopCred = new Shop();
        shopCred.setCredit(shopCred.getCredit() - 75);
        System.out.println("A " + shopCred.getCredit()
            + "charge is added to your
account. Thank you for your business!");
    }

    }
    } else if (response2 == 3) {
        Bird.birdBuy(birdStock);
        String chosenBird = scnr.next().trim();
        Iterator<Bird> itr = birdStock.iterator();
        while (itr.hasNext()) {
            Bird tempBird = itr.next();
            if
(tempBird.getBirdType().equalsIgnoreCase(chosenBird)) {
                itr.remove();
                animalCount.remove(tempBird.getId());

Animal.setNumAnimals(Animal.getNumAnimals() - 1);
                System.out.println("Thank you for your
business!");
            }
        }
    } else if (response2 == 2) {
        Reptile.reptileBuy(reptileStock);
        String chosenReptile = scnr.next().trim();
        Iterator<Reptile> itr = reptileStock.iterator();
        while (itr.hasNext()) {
            Reptile tempRep = itr.next();
            if
(tempRep.getReptileType().equalsIgnoreCase(chosenReptile)) {
                itr.remove();
                animalCount.remove(tempRep.getId());

Animal.setNumAnimals(Animal.getNumAnimals() - 1);
                Shop shopCred = new Shop();

```

```

shopCred.setCredit(shopCred.getCredit() - 25);
System.out.println("You now have a $" +
shopCred.getCredit() + " charge to your account");
System.out.println("Thank you for your
business!");
    }
}
} else if (response.equalsIgnoreCase("Reptile")) {
    System.out.println("What type of reptile are you bringing
in?");

    String reptileType = scnr.next();
    Reptile reptile5 = new Reptile();
    System.out.println("What is the age, weight, and color of
your reptile?");

    int rAge = scnr.nextInt();
    double rWeight = scnr.nextDouble();
    String rColor = scnr.next();
    reptile5.setReptileType(reptileType);
    reptile5.setAge(rAge);
    reptile5.setWeight(bWeight);
    reptile5.setScalesColor(bColor);
    reptile5.setPrice(100);
    reptile5.setId(Animal.getNumAnimals());
    reptileStock.add(reptile5);
    animalCount.put(reptile5.getId(), reptile5);
    System.out.println("reptiles are worth $50");
    System.out.println("Woud you like to exchange for:");
    System.out.println("1. Mammal and pay $50");
    System.out.println("2. Reptile and pay/receive no credit");
    System.out.println("3. Bird and receive $25");
    int response3 = scnr.nextInt();
    if (response3 == 1) {
        Mammal.mammalBuy(mammalStock);
        String chosenMammal = scnr.next().trim();
        Iterator<Mammal> itr = mammalStock.iterator();
        while (itr.hasNext()) {
            Mammal tempMam = itr.next();
            if
(tempMam.getMammalType().equalsIgnoreCase(chosenMammal)) {
                itr.remove();

```

```

        animalCount.remove(tempMam.getId());
    }
    Animal.setNumAnimals(Animal.getNumAnimals() - 1);
    Shop shopCred = new Shop();
    shopCred.setCredit(shopCred.getCredit() - 50);
    System.out.println("A " +
        shopCred.getCredit()
        + "charge is added to your
    account. Thank you for your business!");
    }
    }
    } else if (response3 == 3) {
        Bird.birdBuy(birdStock);
        String chosenBird = scnr.next().trim();
        Iterator<Bird> itr = birdStock.iterator();
        while (itr.hasNext()) {
            Bird tempBird = itr.next();
            if
(tempBird.getBirdType().equalsIgnoreCase(chosenBird)) {
                itr.remove();
                animalCount.remove(tempBird.getId());
            }
            Animal.setNumAnimals(Animal.getNumAnimals() - 1);
            Shop shopCred = new Shop();
            shopCred.setCredit(25);
            System.out.println(
                "You have received a $" +
                shopCred.getCredit() + " credit to your account");
            System.out.println("Thank you for your
            business!");
        }
    }
    } else if (response3 == 2) {
        Reptile.reptileBuy(reptileStock);
        String chosenReptile = scnr.next().trim();
        Iterator<Reptile> itr = reptileStock.iterator();
        while (itr.hasNext()) {
            Reptile tempRep = itr.next();

```

```

                                if
(tempRep.getReptileType().equalsIgnoreCase(chosenReptile)) {
                                itr.remove();
                                animalCount.remove(tempRep.getId());

Animal.setNumAnimals(Animal.getNumAnimals() - 1);
                                System.out.println("Thank you for your
business!");
                                }
                                }
                                }
                                }
                                }
                                }

public void welcome() {
    Shop shop2 = new Shop();
    System.out.println("Welcome to the Pet Shop! Are you here to: ");
    System.out.println("1. Adopt");
    System.out.println("2. Sell");
    System.out.println("3. Exchange");
    int choice = scnr.nextInt();
    if (choice == 1) {
        shop2.initiateAdopt(scnr);
    } else if (choice == 2) {
        shop2.initiateSell(scnr);
    } else if (choice == 3) {
        shop2.initiateExchange(scnr);
    } else {
        System.out.println("thanks for stopping by!");
    }
}

public static void main(String[] args) {

    Scanner scnr = new Scanner(System.in);
    Shop shop = new Shop();
    Shop shop2 = new Shop();
    shop2.populateMammal();
    shop2.populateBird();

```

```

shop2.populateReptile();

shop2.welcome();

System.out.println(Animal.getNumAnimals());
}
}

```



Buy Scenario

- Prompted with the option buy
- A list of types of animals is prompted to buy (Reptile, Bird, Mammal)
- The user then chooses what type of animal to buy
 - Mammal
 - The user is given the option to buy between gerbil, dog, cat
 - Each mammal shows age, weight, and fur type
 - Bird
 - The user is given the option to buy between cardinal, parrot, blue jay, and parakeet.
 - Each bird shows age, weight, and
 - Reptile
 - The user is given the option to buy between Gecko, Snake, Komodo Dragon
- After the animal is decided by the user, that animal is taken out of that specific animal class (HashMap) and animal count(ArrayList) from the stock itself
- The User then is prompted to a price range of high, medium and low price (mammal is 100, reptile is 50, bird 25)
- The user then has the option to buy or deny the adoption of the animal.
 - If accepted the user is prompted with “congratulated with their new pet”
 - If denied the user is prompted with “sorry come again”

Sell Scenario

- The user is prompted "what animal are you willing to sell" between three options of mammal, bird, reptile
- If picked mammal
 - The user is asked what specific mammal they are bringing
 - Then the user is prompted with a price of \$100 since it is a mammal
- If picked bird
 - The user is asked what specific bird they are bringing
 - The user is prompted with a price of \$25 since it is a bird
- If picked reptile
 - The user is asked what specific reptile they are bringing
 - The user is prompted with a price of \$50 since it is a reptile
- The user is then prompted to accept or deny
 - If accepted the console prints "Thank you for your business"
 - If denied the console prints "sorry come again"

Exchange scenario

- User prompted "What animal are you bringing in?"
- Reptile, bird, and mammal displayed
 - If user types reptile, prompted to input reptile type, then age weight and color
 - Will then display price of reptile (50) and will prompt with 3 exchange options:
 - A mammal and pay \$50
 - A bird and receive \$25
 - A reptile and no credit
 - If user types bird, prompted to input bird type, then age weight and color
 - Will then display price of bird (25) and will prompt with 3 exchange options:
 - A mammal and pay \$75
 - A bird and no credit
 - A reptile and pay \$25

- If user types mammal, prompted to input mammal type, then age weight and color
 - Will then display price of mammal (100) and will prompt with 3 exchange options:
 - A mammal and no credit
 - A bird and receive \$75
 - A reptile and receive \$50